

Preliminary Research Document

Agentic Customer 360 — Proactive Intervention Desk
Inter IIT Tech Meet 15.0 Prepathon (NLP), IIT (BHU)

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References Consulted

1. Akidau et al. — The Dataflow Model: A Practical Approach to Balancing Correctness, Latency, and Cost in Massive-Scale, Unbounded, Out-of-Order Data Processing (VLDB 2015). Decoupling event time from processing time; utilizing sliding watermarks and accumulation buffers to absorb late telemetry without state corruption. *Informs: priority-heap watermark reordering buffer handling asynchronous ingestion across multi-domain banking events.*
2. C. E. Shannon — A Mathematical Theory of Communication (Bell System Technical Journal, 1948). Quantification of information entropy and surprise (ΔH) in stochastic event sequences. *Informs: information-entropy trigger gating that suppresses unneeded LLM agent invocations on routine, low-variance customer transactions, cutting token cost by over 75%.*
3. Zhong et al. — MemoryBank: Enhancing Large Language Models with Long-Term Memory (IJCAI 2023). Formalizing psychological forgetting curves ($e^{-\lambda \Delta t}$) to continuously decay historical behavioral memory strength. *Informs: exponential state decay ($t_{1/2} = 30$ days) on the customer state board so transient churn markers naturally attenuate over time.*
4. Du et al. — Improving Factuality and Reasoning in Language Models through Multi-Agent Debate (MIT / Google DeepMind, 2023). Independent agents argue divergent hypotheses over multiple rounds, forcing fact-grounded resolution rather than averaging uncertainty. *Informs: agent debate module resolving explicit contradictions between the Usage Agent (churn risk) and Transaction Agent (wealth windfall).*
5. Madaan et al. — Self-Refine: Iterative Refinement with Self-Feedback (NeurIPS 2023). Separating initial draft generation from multi-attribute critical review against strict rubrics. *Informs: Critique-Refiner Agent validating proposed interventions against customer-tier monetary caps and tone policies before human routing.*
6. H. P. Nii — The Blackboard Model of Problem Solving and the Blackboard Framework (AI Magazine, 1986). Independent specialized knowledge sources cooperating via a structured, shared working state without direct point-to-point coupling. *Informs: Customer Shared State Board where parallel swarm agents write discrete telemetry findings rather than passing full prompt traces.*
7. NIST SP 800-53 (Rev. 5) — Security and Privacy Controls for Information Systems (Dual Control / Two-Person Rule). Multi-custody authorization interlocks preventing catastrophic single-point failures during irreversible state modifications. *Informs: Two-Man Blast Shield escrowing high-value actions ($\geq \$500$) until verified via typed friction inputs and dual operator sign-off.*
8. Saltzer & Schroeder — The Protection of Information in Computer Systems (Classic Principles). Fail-safe defaults and economy of mechanism; hard security policies must not depend on complex non-deterministic heuristics. *Informs: deterministic regex guardrail triplines intercepting legal and fraud keywords to bypass LLM autonomy and immediately freeze live communication.*

9. Fellegi & Sunter — A Theory for Record Linkage (JASA 1969). Probabilistic mathematical framework for cross-system entity resolution and matching under noisy identifiers. *Informs: multi-domain evidence accumulator linking anonymized card payments, KYC updates, and web app telemetry into unified life-event anchors.*
10. Lewis et al. — Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (NeurIPS 2020). Non-parametric dense retrieval coupled with parametric generation for grounded fact verification. *Informs: Semantic Policy Memory querying product eligibility matrices and tier-based fee-waiver thresholds with explicit citation IDs.*
11. Ribeiro et al. — "Why Should I Trust You?": Explaining the Predictions of Any Classifier (KDD 2016). First-class explainability requires traceable local evidence citations rather than post-hoc generalized rationalizations. *Informs: citation-backed checkpoint logging where every intervention explicitly references triggering event IDs and active baseline Z-scores.*
12. Blázquez-García et al. — A Review on Outlier/Anomaly Detection in Time Series Data (IEEE TKDE 2021). Adaptive rolling window statistics and standard deviation deviation metrics over non-stationary financial data streams. *Informs: inline spend anomaly scoring ($Z = \frac{x-\mu}{\sigma}$) and rolling 30-day login velocity trend transforms computed on the live stream.*